

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **Borane-N,N-diethylaniline complex**  
 Product Description: **Borane-N,N-diethylaniline complex**  
 Cat No. : 176770000; 176770250; 176771000; 176772500  
 Synonyms N,N-Diethylanilineborane  
 CAS No 13289-97-9  
 Molecular Formula C10 H18 B N

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

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 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                                        |                   |
|------------------------------------------------------------------------|-------------------|
| Substances/mixtures which, in contact with water, emit flammable gases | Category 1 (H260) |
| Acute oral toxicity                                                    | Category 3 (H301) |
| Acute dermal toxicity                                                  | Category 3 (H311) |
| Acute Inhalation Toxicity - Vapors                                     | Category 2 (H330) |
| Specific target organ toxicity - (repeated exposure)                   | Category 2 (H373) |
| Chronic aquatic toxicity                                               | Category 2 (H411) |

**Label Elements**


Signal Word

Danger

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## Hazard Statements

H260 - In contact with water releases flammable gases which may ignite spontaneously  
H330 - Fatal if inhaled  
H411 - Toxic to aquatic life with long lasting effects  
H373 - May cause damage to organs through prolonged or repeated exposure  
H301 + H311 - Toxic if swallowed or in contact with skin

## Precautionary Statements

### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P311 - Call a POISON CENTER or doctor  
P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P330 - Rinse mouth  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P361 + P364 - Take off immediately all contaminated clothing and wash it before reuse

### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

EUH014 - Reacts violently with water  
Combustible liquid

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                        | CAS No     | Weight % |
|--------------------------------------------------|------------|----------|
| N,N-Diethylaniline                               | 91-66-7    | <3       |
| Boron, (N,N-diethylbenzenamine)trihydro-, (T-4)- | 13289-97-9 | >95      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

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|                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ingestion</b>                                                                                                                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                                                                                                                          | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration. |
| <b>Self-Protection of the First Aider</b>                                                                                                  | Use personal protective equipment as required.                                                                                                                                                                                                                                                                                 |
| <b><u>Most important symptoms and effects, both acute and delayed</u></b>                                                                  |                                                                                                                                                                                                                                                                                                                                |
| Difficulty in breathing. May cause methemoglobinemia. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. |                                                                                                                                                                                                                                                                                                                                |
| <b><u>Indication of any immediate medical attention and special treatment needed</u></b>                                                   |                                                                                                                                                                                                                                                                                                                                |
| <b>Notes to Physician</b>                                                                                                                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                         |

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

Water.

### Special hazards arising from the substance or mixture

Reacts violently with water. Contact with water liberates extremely flammable gases. Combustible material. Containers may explode when heated. Keep product and empty container away from heat and sources of ignition. Risk of ignition.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Nitrogen oxides (NO<sub>x</sub>), Oxides of boron.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental precautions

Do not flush into surface water or sanitary sewer system.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Do not expose spill to water. Remove all sources of ignition.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

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## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Do not breathe mist/vapors/spray. Do not allow contact with water. Keep away from open flames, hot surfaces and sources of ignition.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from water or moist air. Keep away from heat, sparks and flame.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

##### **Eye Protection**

Wear safety glasses with side shields (or goggles)

##### **Hand Protection**

Protective gloves

##### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

##### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

##### **Recommended Filter type:**

Organic gases and vapours filter Type A Brown conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system

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## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                     |                               |                                   |
|---------------------|-------------------------------|-----------------------------------|
| Appearance          | Colorless, Amber              |                                   |
| Physical State      | Liquid                        |                                   |
| Odor                | No information available      |                                   |
| Odor Threshold      | No data available             |                                   |
| pH                  | No information available      |                                   |
| Melting Point/Range | -30 - -27 °C / -22 - -16.6 °F |                                   |
| Softening Point     | No data available             |                                   |
| Boiling Point/Range | No information available      |                                   |
| Flash Point         | 63 °C / 145.4 °F              | Method - No information available |

|                          |                          |        |
|--------------------------|--------------------------|--------|
| Evaporation Rate         | No information available |        |
| Flammability (solid,gas) | Not applicable           | Liquid |
| Explosion Limits         | No data available        |        |

|                              |                             |             |
|------------------------------|-----------------------------|-------------|
| Vapor Pressure               | No information available    |             |
| Vapor Density                | 5.62                        | (Air = 1.0) |
| Specific Gravity / Density   | 0.920                       |             |
| Bulk Density                 | Not applicable              | Liquid      |
| Water Solubility             | Reacts violently with water |             |
| Solubility in other solvents | No information available    |             |

### Partition Coefficient (n-octanol/water)

|                    |         |
|--------------------|---------|
| Component          | log Pow |
| N,N-Diethylaniline | 3.31    |

|                           |                          |                                        |
|---------------------------|--------------------------|----------------------------------------|
| Autoignition Temperature  | No data available        |                                        |
| Decomposition Temperature | No data available        |                                        |
| Viscosity                 | No data available        |                                        |
| Explosive Properties      |                          | explosive air/vapour mixtures possible |
| Oxidizing Properties      | No information available |                                        |

|                   |             |
|-------------------|-------------|
| Molecular Formula | C10 H18 B N |
| Molecular Weight  | 163.07      |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

Yes.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

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## Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.  
None under normal processing. Reacts violently with water.

## Conditions to Avoid

Incompatible products. Excess heat. Exposure to moist air or water. Exposure to moisture.  
Keep away from open flames, hot surfaces and sources of ignition.

## Incompatible Materials

Strong oxidizing agents. Water. Acids. Alcohols. Bases.

## Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Oxides of boron.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

#### (a) acute toxicity;

|            |            |
|------------|------------|
| Oral       | Category 3 |
| Dermal     | Category 3 |
| Inhalation | Category 2 |

| Component          | LD50 Oral                | LD50 Dermal         | LC50 Inhalation                           |
|--------------------|--------------------------|---------------------|-------------------------------------------|
| N,N-Diethylaniline | LD50 = 606 mg/kg ( Rat ) | >5000 mg/kg ( Rat ) | LC50 = 1920 mg/m <sup>3</sup> ( Rat ) 4 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

#### (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; Category 2

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**Route of exposure  
Target Organs**

Oral Dermal, Inhalation,  
Blood.

**(j) aspiration hazard;**

No data available

**Other Adverse Effects**

The toxicological properties have not been fully investigated.

**Symptoms / effects, both acute and delayed**

May cause methemoglobinemia. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

**Endocrine Disrupting Properties**

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects**

The product contains following substances which are hazardous for the environment. Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. Reacts with water so no ecotoxicity data for the substance is available.

| Component          | Freshwater Fish                                                                                                 | Water Flea                                               | Freshwater Algae | Microtox                                          |
|--------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------|---------------------------------------------------|
| N,N-Diethylaniline | LC50: = 16.4 mg/L, 96h flow-through<br>(Pimephales promelas)<br>LC50: = 38.5 mg/L, 96h<br>(Oncorhynchus mykiss) | EC50: 1.0 - 1.6 mg/L, 48h semi-static<br>(Daphnia magna) |                  | EC50 = 6.50 mg/L 5 min<br>EC50 = 7.70 mg/L 15 min |

**Persistence and degradability**

**Persistence  
Degradability  
Degradation in sewage treatment plant**

No information available  
Persistence is unlikely, based on information available.  
Reacts with water.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants. Reacts violently with water.

**Bioaccumulative potential**

Product does not bioaccumulate due to reaction with water

| Component          | log Pow | Bioconcentration factor (BCF) |
|--------------------|---------|-------------------------------|
| N,N-Diethylaniline | 3.31    | 17 - 125 dimensionless        |

**Mobility in soil**

Reacts violently with water. . Is not likely mobile in the environment.

**Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

**Other adverse effects**

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**

**Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

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## Other Information

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Can be landfilled or incinerated, when in compliance with local regulations Do not let this chemical enter the environment Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN3148  
Hazard Class 4.3  
Packing Group I  
Proper Shipping Name Water reactive liquid, n.o.s Boron, (N,N-diethylbenzenamine)trihydro-, (T-4)-, N,N-Diethylaniline

### Road and Rail Transport

UN-No UN3148  
Hazard Class 4.3  
Packing Group I  
Proper Shipping Name Water reactive liquid, n.o.s Boron, (N,N-diethylbenzenamine)trihydro-, (T-4)-, N,N-Diethylaniline

### IATA

UN-No UN3148  
Hazard Class 4.3  
Packing Group I  
Proper Shipping Name Water reactive liquid, n.o.s Boron, (N,N-diethylbenzenamine)trihydro-, (T-4)-, N,N-Diethylaniline

## Special Precautions for User

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

## International Inventories

X = listed

| Component                                        | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|--------------------------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| N,N-Diethylaniline                               | 202-088-8 | X    | X   | X     | X    | X    | X     | X    | KE-10434 |
| Boron, (N,N-diethylbenzenamine)trihydro-, (T-4)- | 236-305-2 | X    | -   | -     | -    | X    | -     | -    | -        |

## National Regulations

### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION



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## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date

21-Mar-2025

Revision Summary

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**